STD 8 - BASIC SCIENCE - UNIT 4: TREE OF LIFE

(Model Question Paper)

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Total Marks: 20

SECTION A

(Questions 1 and 2 carry 1 mark each) (2 x 1 = 2)

1. Analyse the statements i) to iv) related to the History of Classification and choose the correct answer from the options given below.

i) Aristotle classified living beings as those with red blood and those without.

ii) John Ray was the first to use the term 'species'.

iii) Ernst Haeckel introduced the Five Kingdom Classification system.

iv) The smallest unit of classification is the Genus.

a. i, ii correct, iii, iv incorrect

b. i, ii, iii correct, iv incorrect

c. i, ii correct, iii, iv incorrect

d. i, iii, iv correct, ii incorrect

2. Choose and write the correct answer from the options given below where the pairs are correctly matched based on the Five Kingdom Classification.

I. Kingdom	**II. Characteristic Feature**
P. Monera	i) Includes unicellular eukaryotes
Q. Protista	ii) Have a cell wall but lack chloroplasts
R. Fungi	iii) Includes multicellular organisms with tissues
S. Animalia	iv) Includes prokaryotic organisms

	(P)	**(Q)**	**(R)**	**(S)**
a.	(i)	(iv)	(iii)	(ii)
b.	(iv)	(i)	(ii)	(iii)
c.	(iii)	(ii)	(i)	(iv)
d.	(iv)	(ii)	(i)	(iii)

SECTION B

(Questions 3 to 8 carry 2 marks each) (6 X 2 = 12)

3. State the contributions of the following two scientists to the history of classification.

i) Theophrastus (1 mark) ii) Carl Linnaeus (1 mark)

4. The Kingdom Protista includes organisms that are different from Monera and Animalia.

i) What is the fundamental difference between the cells of Monera and Protista? (1 mark)

ii) Give one example of an organism belonging to the Kingdom Protista. (1 mark)

5. Observe the following scientific name of the Peacock: *Pavo cristatus*

i) According to the rules of Binomial Nomenclature, which part of the name (*Pavo* or *cristatus*) represents the species? (1 mark)

ii) What is the significance of the Genus name in the classification hierarchy? (1 mark)

6. The smallest unit of the classification hierarchy is the species.

i) Define what a species is. (1 mark)

ii) State any one characteristic of the organisms belonging to the Kingdom Monera. (1 mark)

7. Carl Linnaeus is known as the Father of Taxonomy.

i) What is meant by Taxonomy? (1 mark)

ii) Write any one rule that must be followed when writing the scientific name of an organism (Binomial Nomenclature). (1 mark)

8. R.H. Whittaker's classification system is widely accepted.

i) Which kingdom is characterised by non-motile, multicellular organisms that contain chloroplasts and have a cell wall? (1 mark)

ii) Give one example of an organism from the Kingdom Fungi. (1 mark)

SECTION C

(Questions 9 and 10 carry 3 marks each) (2 X 3 = 6)

9. The process of arranging organisms into different groups and categories is essential for biological study.

i) State the full sequence of the Taxonomic Hierarchy developed by Carl Linnaeus, starting from the largest unit. (1 mark)

ii) Explain the significance of this hierarchy in classifying organisms. (2 marks)

10. (A). The Five Kingdom Classification System was introduced by R. H. Whittaker in 1969. Explain the two different kingdoms that organisms can belong to if they are:

- Unicellular (1.5 marks)

- Multicellular with tissues (1.5 marks) *(For each category, state the name of the relevant kingdom and one distinct characteristic of the organisms belonging to it.)*

OR

10. (B). Binomial Nomenclature is the scientific method of naming organisms.

i) Who developed this method, and why is it considered scientific? (1 mark)

ii) Explain the two main components (names) used in this naming system with an example. (2 marks)